Exercises 15 and 16 use the notation of Example 1 for matrices in echelon form. Suppose each matrix represents the augmented matrix for a system of linear equations. In each case, determine if the system is consistent. If the system is consistent, determine if the solution is unique.

15) a) $\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & \square & 0 \end{bmatrix}$

The corresponding equations are:

1. $a_1x + b_1y + c_1z = d_1$
2. $b_2y + c_2z = d_2$
3. $c_3z = 0$

$a_1 \neq 0, b_2 \neq 0, c_3 \neq 0$. From eqn 3, we get a unique value of z.
Substituting value of z in eqn 2, we get a unique value of y.
Substituting value of z and y in eqn 1, we get a unique value of x.

$\boxed{\therefore \text{The system is consistent and the solution is unique}}$

b) $\begin{bmatrix} 0 & \square & * & * & * \\ 0 & 0 & \square & * & * \\ 0 & 0 & 0 & 0 & \square \end{bmatrix}$

The corresponding equations are:

1. $b_1y + c_1z + d_1w = e_1$
2. $c_2z + d_2w = e_2$
3. $0 = e_3$

$b_1 \neq 0$ $c_2 \neq 0$ $e_3 \neq 0$. By Theorem 2, due to the presence of eqn 3, we conclude that the $\boxed{\text{system is inconsistent}}$.

16) a) $\begin{bmatrix} \square & * & * \\ 0 & \square & * \\ 0 & 0 & 0 \end{bmatrix}$

The corresponding equations are:

1. $a_1x_1 + b_1x_2 = c_1$ $\quad a_1 \neq 0$
2. $b_2x_2 = c_2$ $\quad b_2 \neq 0$
3. $0 = 0$

From eqn 2, we get a unique value of x_2. Substituting the value of x_2 in eqn 1, we get a unique value of x_1.

$\boxed{\therefore \text{The system is consistent and the solution is unique.}}$